

SHOPEE BARCODE & VIDEO MANAGEMENT SYSTEM

System Design Document: ERD & Architecture Diagrams

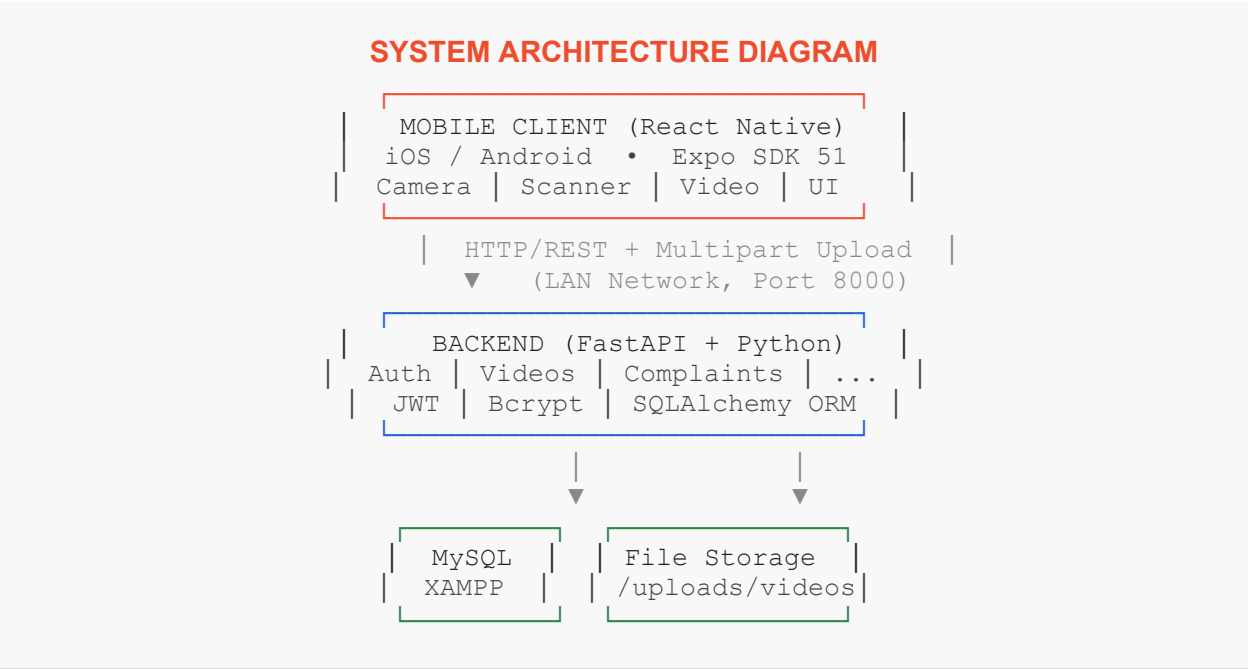
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Document	System Design Document	Version	1.0
Project	Shopee Barcode Video	Date	May 2025
Prepared By	Mẫn Ngọc Đắc	Status	Final

1. System Architecture

1.1 Architecture Overview

The Shopee Barcode & Video Management System follows a client-server architecture with three distinct layers: the mobile frontend, the REST API backend, and the database/storage layer. The system is designed for local network deployment, suitable for warehouse or store environments.



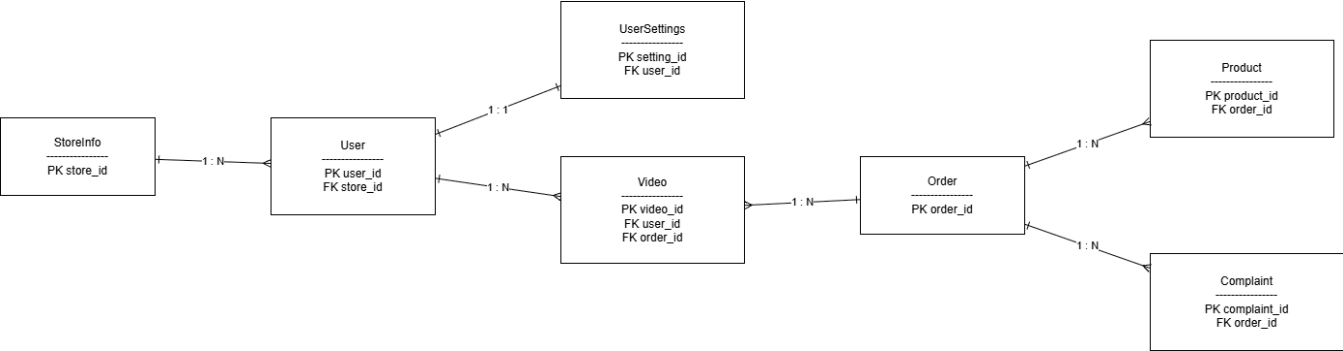
1.2 Component Responsibilities

Component	Technology	Responsibilities
Mobile App	React Native, Expo SDK 51, TypeScript	UI rendering, camera control, QR scanning, video recording, API calls, local auth storage
API Server	Python 3.13, FastAPI, Uvicorn	REST endpoints, JWT auth, business logic, file handling, database operations
ORM Layer	SQLAlchemy	Database abstraction, model definitions, relationship mapping
Database	MySQL (XAMPP)	Persistent storage of users, orders, videos, complaints, settings
File Storage	Local Filesystem	Storing uploaded MP4 video files under /uploads/videos/
Auth Module	JWT (python-jose), bcrypt (passlib)	Token generation/validation, password hashing/verification

2. Entity Relationship Diagram (ERD)

2.1 ERD Overview

The database consists of 7 entities. The diagram below illustrates the relationships between entities. Primary keys are shown in orange, foreign keys are indicated in the constraint column.



2.2 Relationship Summary

Entity A	Relationship	Entity B	Foreign Key
StoreInfo	One-to-Many	User	User.store_id
User	One-to-One	UserSettings	UserSettings.user_id
User	One-to-Many	Video	Video.user_id
Order	One-to-Many	Video	Video.order_id
Order	One-to-Many	Product	Product.order_id
Order	One-to-Many	Complaint	Complaint.order_id

3. Database Schema — Entity Attribute Tables

3.1 User

Stores user account credentials, role, and store assignment.

users			
Column Name	Data Type	Constraint	Description
user_id	VARCHAR(36)	PK, NOT NULL	UUID — unique user identifier
name	VARCHAR(50)	NOT NULL	Full name of the user
email	VARCHAR(100)	UNIQUE, NOT NULL	Login email address
password	VARCHAR(255)	NOT NULL	bcrypt-hashed password
role	ENUM	NOT NULL, DEFAULT employee	Role: employee or admin
store_id	INTEGER	FK -> store_info.id, NULLABLE	Assigned store (optional)
created_at	DATETIME	DEFAULT utcnow	Account creation timestamp
updated_at	DATETIME	DEFAULT utcnow, ON UPDATE	Last modification timestamp

3.2 Order

Stores Shopee order information linked to videos and complaints.

orders			
Column Name	Data Type	Constraint	Description
order_id	VARCHAR(512)	PK, NOT NULL	Shopee order ID (may be long, hashed if >255 chars)
customer_name	VARCHAR(100)	NULLABLE	Customer full name
customer_address	VARCHAR(512)	NULLABLE	Delivery address
order_date	DATETIME	DEFAULT utcnow	Date the order was placed
status	ENUM	DEFAULT pending	Status: pending, packing, or completed
created_at	DATETIME	DEFAULT utcnow	Record creation timestamp
updated_at	DATETIME	ON UPDATE utcnow	Last update timestamp

3.3 Video

Stores video recording metadata, linked to an order and the recording user.

videos			
Column Name	Data Type	Constraint	Description
video_id	VARCHAR(36)	PK, NOT NULL	UUID — unique video record identifier
order_id	VARCHAR(512)	FK -> orders.order_id, NOT NULL	Associated Shopee order
user_id	VARCHAR(36)	FK -> users.user_id, NULLABLE	Employee who recorded the video
file_name	VARCHAR(512)	NOT NULL	Stored file name on server (sanitized)
storage_url	VARCHAR(512)	NULLABLE	Relative URL path to the video file
start_time	DATETIME	NULLABLE	Recording start timestamp
end_time	DATETIME	NULLABLE	Recording end / upload timestamp
created_at	DATETIME	DEFAULT utcnow	Record creation timestamp
updated_at	DATETIME	ON UPDATE utcnow	Last update timestamp

3.4 Complaint

Tracks order-related complaints submitted by employees and managed by admins.

complaints			
Column Name	Data Type	Constraint	Description
complaint_id	VARCHAR(36)	PK, NOT NULL	UUID — unique complaint identifier
order_id	VARCHAR(36)	FK -> orders.order_id, NOT NULL	The order this complaint relates to
issue_type	ENUM	NOT NULL	Type: missing_item, wrong_item, or other
description	TEXT	NOT NULL	Detailed description of the issue
resolved	BOOLEAN	DEFAULT false	Whether the complaint has been resolved
created_at	DATETIME	DEFAULT utcnow	Complaint submission timestamp

resolved_at	DATETIME	NULLABLE	Timestamp when complaint was resolved
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3.5 Product

Stores product items associated with an order, including barcode information.

products			
Column Name	Data Type	Constraint	Description
product_id	VARCHAR (255)	PK, NOT NULL	Unique product identifier
order_id	VARCHAR (255)	FK -> orders.order_id	Associated order
name	VARCHAR (100)	NULLABLE	Product name
quantity	INTEGER	NULLABLE	Quantity in order
barcode	VARCHAR (100)	NULLABLE	Product barcode string
video_url	VARCHAR (512)	NULLABLE	Optional video URL for this product
created_at	DATETIME	DEFAULT utcnow	Record creation timestamp
updated_at	DATETIME	ON UPDATE utcnow	Last update timestamp

3.6 UserSettings

Stores per-user application preferences (dark mode, language).

user_settings			
Column Name	Data Type	Constraint	Description
user_id	VARCHAR (36)	PK, FK -> users.user_id	One-to-one with User entity
dark_mode	BOOLEAN	DEFAULT false	Dark mode preference toggle
language	VARCHAR (20)	DEFAULT 'en'	Preferred UI language code
created_at	DATETIME	DEFAULT utcnow	Settings creation timestamp
updated_at	DATETIME	ON UPDATE utcnow	Last settings update timestamp

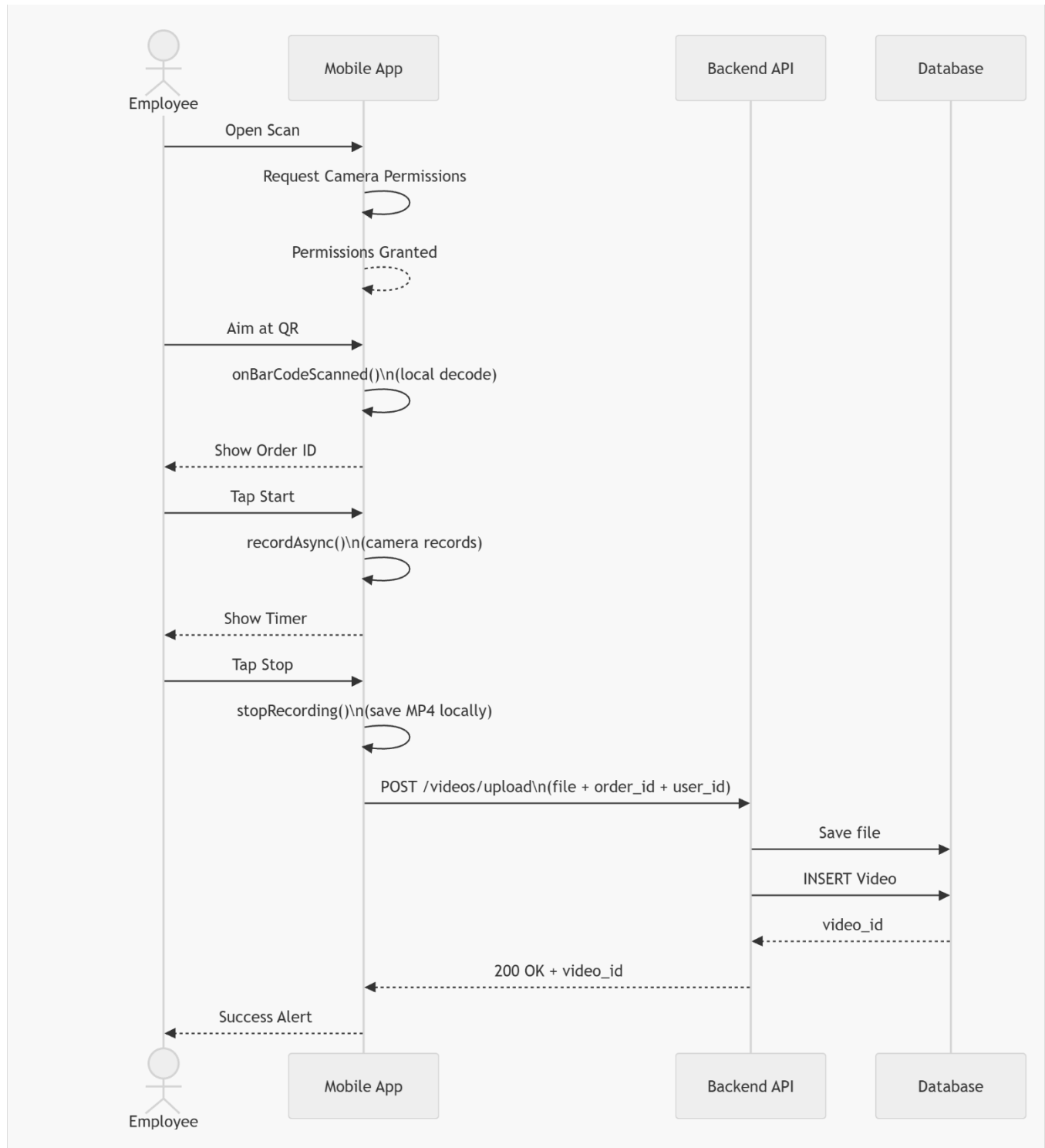
3.7 StoreInfo

Stores information about the physical store or warehouse location.

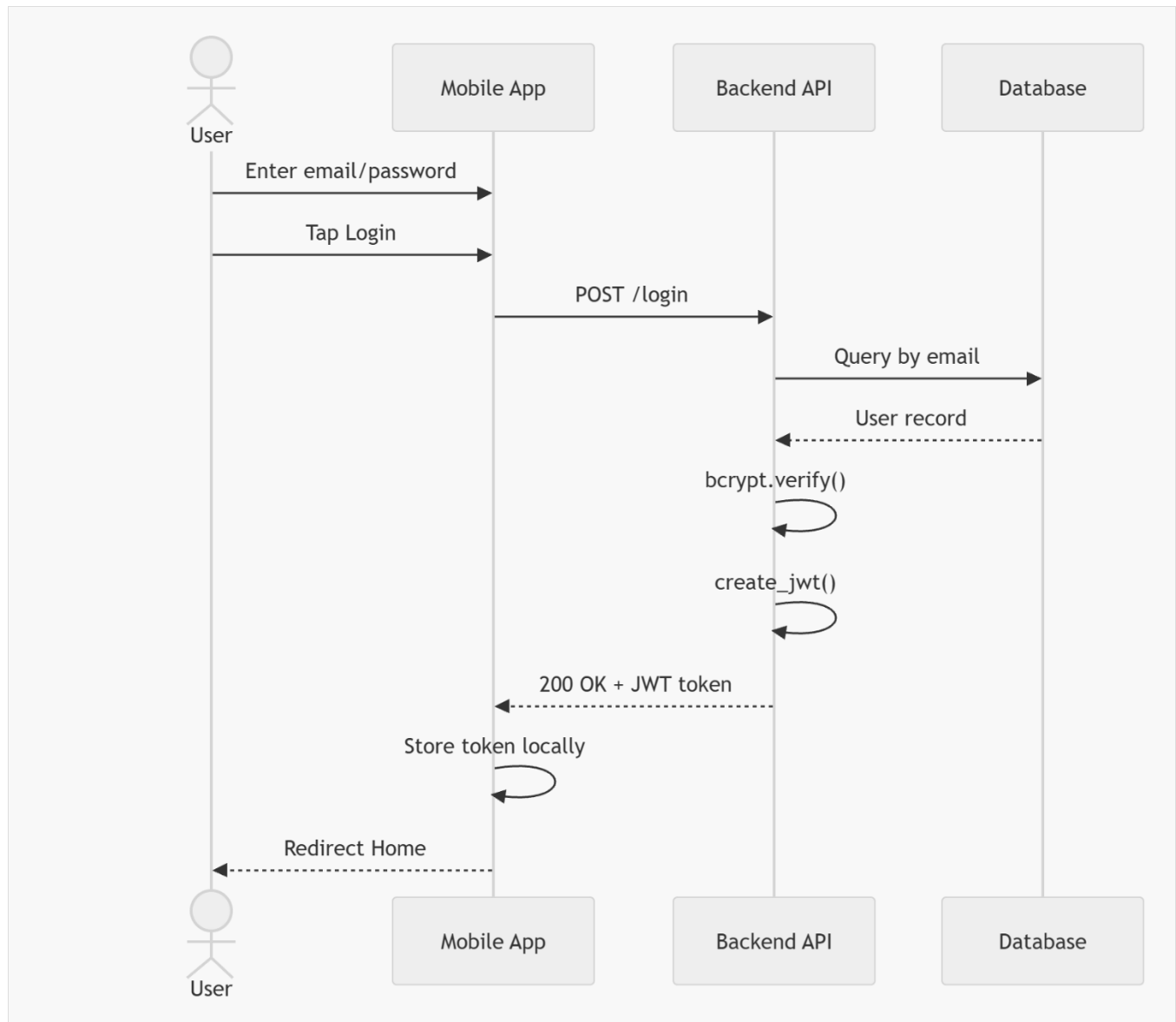
store_info			
Column Name	Data Type	Constraint	Description
id	INTEGER	PK, AUTO INCREMENT	Unique store identifier
name	VARCHAR (255)	NOT NULL	Store display name
email	VARCHAR (255)	NULLABLE	Store contact email
phone	VARCHAR (50)	NULLABLE	Store contact phone number
location	VARCHAR (255)	NULLABLE	Physical address or location
created_at	DATETIME	DEFAULT utcnow	Record creation timestamp
updated_at	DATETIME	ON UPDATE utcnow	Last update timestamp

4. Key Process Flows

4.1 Scan & Record Video Flow

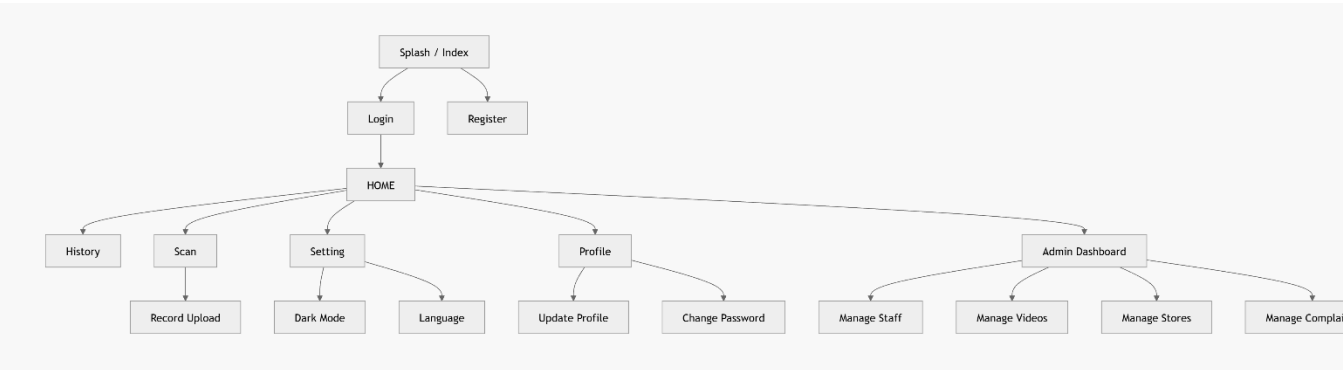


4.2 Authentication Flow



5. Screen Navigation Map

The following diagram shows the navigation structure of the mobile application:



6. Revision History

Version	Date	Author	Description
1.0	May 2025	Mẫn Ngọc Đặc	Initial System Design Document created